Circuit Simulation Lab:15

Design of Asynchronous Circuits

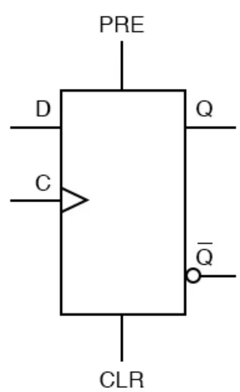
Introduction

The purpose of this experiment is to introduce the design of asynchronous sequential circuits, in this case Asyn-D Flipflop. Asynchronous inputs on a flip-flop have control over the outputs (Q and not-Q) regardless of clock input status. These inputs are called the preset (PRE) and clear (CLR). The preset input drives the flip-flop to a set state while the clear input drives it to a reset state.

Software tools Requirement

Modelsim (Siemens)

Xilinx Vivado



Asynchronous D Flip Flop

Describe the above circuits in Verilog HDL and capture the Waveforms

1) **Behavior modeling**